

# QUI LOC PHAN

🏠 Konatus

✉ qui-loc.phan@konatus.io

👤 locphan2001.github.io

🌐 Vietnamese

## Education

### Université Paris Dauphine & École des Mines de Paris

2023 – 2025

*Master in Computer Science*

*Paris, France*

- Graduated with highest grade in Combinatorial Optimization specialization of program
- Awarded Excellence Scholarship from partner institution ENS Paris Saclay and ENS Paris for 2 consecutive years

### Ho Chi Minh University of Education

2019 – 2023

*Honours Bachelor of Mathematics*

*Ho Chi Minh, Vietnam*

- Graduated with Valedictorian Achievement (rank 1st/150)

## Industrial Experience

### Konatus, Startup of École Polytechnique

Sep 2025 – now

*Research Engineer*

*Palaiseau, France*

- Research on scheduling and resource allocation problems for large-scale corporate customers with complex projects
- Improve internal planning tools using evolutionary computation and machine learning to enhance decision efficiency
- Strengthen robustness and flexibility of solution by modeling uncertainty, adapting to resource variability and disruptions
- Evaluate and benchmark the company's solutions under different uncertainty models and operating conditions
- Collaborate with academic researchers to keep methods in line with cutting-edge techniques in operations research
- Work hand-in-hand with industry experts to deploy and validate methods on real historical and operational datasets

### Corporate Research & Sciences Department, Dassault Systèmes

Mar 2025 – Sep 2025

*Research Engineer Intern*

*Vélizy-Villacoublay, France*

- Study quantum algorithms for constraint programming: optimization problems with integer and logical constraints
- Develop and implement for practical use cases in long-term, explore near-term challenges, conduct comparative analysis
- Practical case study on Healthcare Staff Scheduling at Paris Hospital with multi-objectives and multi-skills

## Academic Experience

### Optimization and Control Group, UMA, ENSTA Paris

May 2024 – Sep 2024

*Graduate Research Intern*

*Palaiseau, France*

- Study how to optimize resources used in quantum circuit with two objectives: number of gates and depth of circuit
- Understand and construct mathematical programming model and graph optimization model for the problems
- Design heuristics algorithms that produce good solutions and benchmark the results with available open-source libraries

### HCMUE & Vietnam Academy of Science and Technology

10 months

*Undergraduate Research Intern*

*Ho Chi Minh, Vietnam*

- Study fundamental aspects of combinatorics and abstract algebra, model and address combinatorial problems
- Design or develop, and evaluate algebraic algorithms both in theory and practice mathematically

## Project

### Production Problem

2024

- Construct mathematical programming model for the resource allocation problem, inspired from the Flowshop Scheduling
- Implement Campbell - Dudek - Smith algorithm, with variant on preprocessing stage and postprocessing stage
- Perform clustering algorithms on preprocessing and local search methods on postprocessing to improve the solution

### Quantum Optimization

2023

- Encode the mathematical programming model of Max-Cut and Maximum Independent Set into quantum computers
- Code end-to-end QAOA algorithms for the problem, test performance with different number of layers and parameters
- Benchmark with evolutionary algorithms on cost function and introduce some heuristic function for better solutions

## Technical Skill

**Language:** Vietnamese (Native), English (C1), French (B1)

**Programming Language:** Python (3 years), C++ (2 years), Matlab (2 years)

**Optimization Solvers:** Gurobi, CPLEX, Google OR-Tools